Distributional regularisers under JEPA bottlenecks: nuisance leakage, dimension and noise on data with known factors

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Abstract Joint-embedding predictive architectures (JEPAs) avoid representational collapse either with an exponential-moving-average (EMA) target or with a regulariser that pulls the embedding distribution towards a target law, typically an isotropic Gaussian. We ask how these regularisers — kernel Stein discrepancies (KSD) above all — behave when two views share a content signal but carry independent nuisance, and how the answer depends on two bottlenecks: the embedding dimension d and Gaussian noise injected into the embedding. On 3DShapes, whose six generative factors are known, we split the factors into content ($k = 6$ probe dimensions) and nuisance, control how often nuisance is resampled between views, and measure nuisance leakage as the held-out R^2 of a linear probe. We compare eight arms (no regulariser, VICReg, SIGReg, sliced and full KSD, MMD, EMA, and a rank-k oracle target) with matched tuning and 5 held-out seeds, plus a temporal tier, a CIFAR-10 tier and an autoencoder tier (747 training runs). All five hypotheses about the JEPA arms are refuted, and the autoencoder hypothesis is confirmed. Isotropic-Gaussian regularisers do not leak more nuisance than VICReg when each is tuned for content; leakage is instead governed by the regularisation weight, and rises with it for every regulariser. Shrinking d towards k reduces leakage for every arm, while injected noise never does. KSD with a prior matched to a non-Gaussian content law does not improve identifiability; full KSD towards $\mathcal{N}(0, I)$ has the best identifiability at every content law. In the temporal tier, EMA alone leaks more than sliced KSD alone. Autoencoders leak more nuisance than every JEPA arm.

1 Introduction

A joint-embedding predictive architecture (JEPA) encodes two related inputs and trains a predictor to map one embedding to the other [1], [2]. The objective has a trivial minimum — a constant embedding — so every JEPA needs a second ingredient. One family uses an exponential-moving-average (EMA) target encoder with a stop-gradient, as in I-JEPA, V-JEPA and V-JEPA 2 [1], [2], [3]. Another adds a regulariser on the batch of embeddings: VICReg constrains variances and covariances [4]; LeJEPA's SIGReg tests one-dimensional projections against a standard Gaussian [5]; KerJEPA uses kernel discrepancies, MMD and kernelised Stein discrepancies (KSD), sliced and unsliced [6]. The Gaussian target is motivated by identifiability: Klindt et al. show that, within their model class, linear identifiability of JEPA latents holds only for Gaussian latents [7].

The identifiability analysis assumes that the two views differ only by noise on the shared latent [7], and none of these JEPA works has ground-truth nuisance in its views. Real view pairs also differ in *nuisance*: lighting, background, pose, augmentation parameters. A distributional regulariser asks the encoder to fill d dimensions with a prescribed spread. If the content occupies fewer than d dimensions, something must fill the rest, and nuisance is available. Whether this happens, and whether the embedding dimension d or an information bottleneck (noise injected into z) controls it, has not been measured, because measuring it needs ground truth for both content and nuisance.

We use 3DShapes [8], a rendered dataset with a full grid of six known factors, to build view pairs whose content is identical and whose nuisance is resampled independently with probability α. Linear probes on frozen embeddings then give content R^2, nuisance-leakage R^2 and an orthogonal identifiability score, each against ground truth. This is an analysis study, not a methods paper: all regularisers we use already exist.

Contributions. Relative to the 35 works we reviewed (Section 2):

1. A controlled measurement of nuisance leakage for JEPA regularisers (VICReg, SIGReg, sliced and full KSD, MMD, EMA, a rank-k oracle) with ground-truth nuisance whose cross-view dependence α is set by the experimenter (Section 4.1, Section 4.4).
2. A joint sweep of the dimensional bottleneck (d below and above the content dimension k) and an information bottleneck (injected noise σ), with the finding that on this data the dimension, not the noise, limits leakage (Section 4.2).
3. A test of whether KSD with a prior matched to a non-Gaussian Euclidean content law improves identifiability, extending a result reported for MMD on compact manifolds [9]; it does not here (Section 4.3).
4. A temporal tier comparing EMA, EMA + sliced KSD and sliced KSD on leakage when nuisance moves frame to frame (Section 4.5), and an autoencoder tier comparing reconstruction with prediction (Section 4.6).
5. A robustness slice showing that the regularisation weight λ, more than the target law, decides the leakage comparison, together with the full protocol, code and per-run results.

All five hypotheses about the JEPA arms (H1–H5), fixed before the sweep, are refuted; we report them as such. The sixth, that autoencoders leak more than JEPAs, is confirmed.

2 Related work

Table 1 condenses the overlap table we built from the reviewed works (35 papers, each verified against arXiv or the venue). Three points shape our claims.

Kernel regularisers for JEPA are not new. KerJEPA already trains JEPAs with sliced and full KSD and MMD, with Gaussian and IMQ kernels and Gaussian and Laplace targets [6]. SPHERE-JEPA uses full MMD and KSD towards the uniform law on the sphere [10], Rectified LpJEPA a rectified generalised Gaussian [11], VISReg a variance + sliced-Wasserstein regulariser [12], and Wang and Isola a hyperspherical uniformity term for contrastive learning [13]. Our KSD and MMD arms are re-implementations of these. KSD itself comes from goodness-of-fit testing [14], [15]; the IMQ kernel controls convergence where Gaussian kernels can fail [16]; sliced KSD is due to Gong et al. [17]; and a recent review covers Stein's method in computational statistics [18]. Stein operators for general multivariate and univariate densities [19], [20], for implicit models (NP-KSD, which we use as a metric) [21], with general operators [22], and Stein bounds for possibly singular normal targets [23] are the background for our matched and oracle targets.

Matched targets and identifiability. Beyond Gaussian Worlds shows that heat-kernel MMD towards a target matched to the latent geometry (sphere, torus) improves linear identifiability on rendered data [9]. Klindt et al. vary the data law under a Gaussian target [7]. Our H3 is therefore a test of whether the matched-target effect carries over to KSD and to Euclidean non-Gaussian content laws on real rendered images. Without supervision or inductive bias, disentangled factors cannot be identified in general [24].

Nuisance and bottlenecks. The only reviewed works that model view-specific nuisance are contrastive (von Kügelgen et al., who prove block-identifiability of content when style is resampled across views [25]) or multi-view information bottleneck methods [26]. Arnez et al. argue, in theory only, that SIGReg with fixed isotropic noise in z is an exact information bottleneck [27]. The VAE family bottlenecks information by a KL weight [28], [29], [30], WAE pushes the aggregate posterior to the prior with MMD [31], and Stein-based VAEs use SVGD [32]. RankMe's effective rank [33] is one of our metrics. Among the reviewed works, none measures nuisance leakage of a JEPA regulariser under controlled nuisance, none sweeps d and injected noise together, and none compares EMA with KSD on leakage in a temporal JEPA; these are the cells we claim. We claim no novelty for the regularisers, for matched targets as an idea, or for the KSD autoencoder.

Table 1: Condensed overlap table (full version with 35 rows and sources in `notes/related_work.md`). "—" means the work does not address the column.

work	regulariser	target	ground-truth nuisance	d vs k	info. bottleneck	identifiability	objective	data
LeJEPA	SIGReg (sliced)	$\mathcal{N}(0, I)$	—	—	—	informal	JEPA	natural
KerJEPA	KSD, MMD (sliced + full)	Gaussian, Laplace	—	—	—	—	JEPA (EMA)	natural
SPHERE-JEPA	MMD, KSD (full)	uniform on sphere	—	—	—	—	JEPA	natural
Beyond Gaussian Worlds	heat-kernel MMD	matched manifold law	—	partial	—	linear	JEPA	rendered
Klindt et al.	SIGReg	$\mathcal{N}(0, I)$	— (noise only)	open question	—	linear (Gaussian)	JEPA	synth. + rendered
Arnez et al.	SIGReg	$\mathcal{N}(0, I)$	—	—	theory only	assumed	temporal JEPA	none
VICReg	var./cov.	moments	implicit	—	—	—	joint emb.	natural
I-/V-JEPA, V-JEPA 2	EMA + stop-grad	—	—	V-JEPA: implicit (narrow predictor)	—	—	(temporal) JEPA	natural
von Kügelgen et al.	InfoNCE-type	—	yes (style)	$d = k$	—	block	contrastive	synth. + rendered
Federici et al.	MIB	—	yes	—	yes	—	multi-view IB	natural
β-VAE, WAE, InfoVAE	KL, MMD	$\mathcal{N}(0, I)$	—	WAE: discussed	yes (β)	metric (β-VAE)	autoencoder	mixed
this study	VICReg, SIGReg, (sliced) KSD, MMD, EMA, oracle	Gaussian; matched uniform/ Laplace; rank-k	**yes, strength α**	**swept**	**swept**	linear (orthogonal)	JEPA, temporal JEPA, AE	rendered + natural

3 Setup

3.1 Data and views

3DShapes (core). 480,000 rendered 64×64 scenes on a full grid of six factors: floor hue, wall hue and object hue (10 values each), scale (8), shape (4) and orientation (15) [8]. We downsample to 32×32 and store the array as a memory-mapped `uint8` file. The **content** s is object shape, scale and object hue; the **nuisance** n is floor hue, wall hue and orientation. For probing, hues are encoded as $(\cos, \sin)$ pairs and shape as a 3-dimensional regular-simplex code, so the content code is $[\cos h, \sin h, \text{scale}, \text{simplex}(\text{shape})]$ with $k = 6$ and the nuisance code has 5 dimensions. A fixed random 10% of the factor grid is held out for probe evaluation and never used in training.

A pair is built by drawing content once, drawing the nuisance of view 1 uniformly, and resampling each nuisance factor of view 2 independently with probability α (Figure 1). At $\alpha = 0$ the two views are the same image; at $\alpha = 1$ their nuisance is independent. Both views receive pixel noise with standard deviation 0.02. The validation gates confirmed that content is identical across views at every α and that at $\alpha = 1$ the largest absolute cross-view correlation of the nuisance codes is 0.0099.

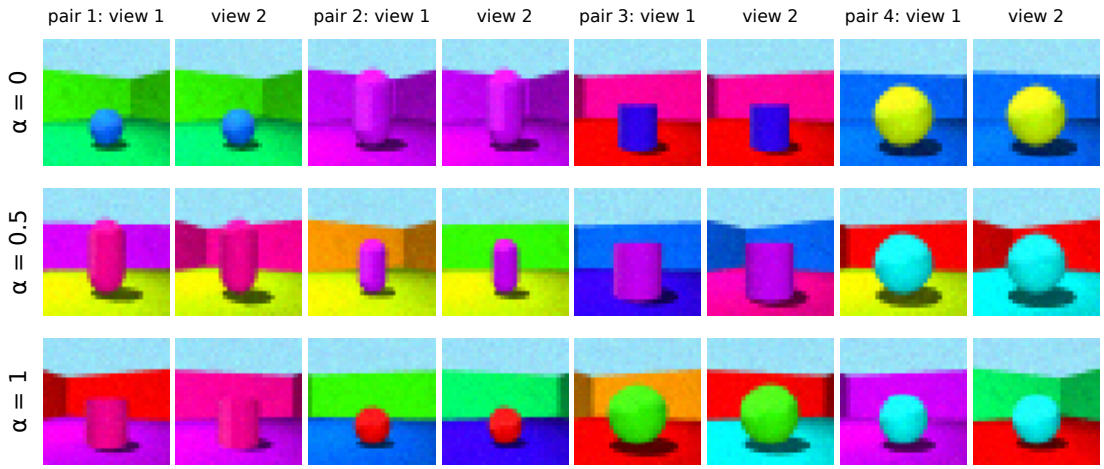


Figure 1: Real 3DShapes view pairs from the pair sampler at three nuisance strengths. Content (object shape, scale, object hue) is shared; floor hue, wall hue and orientation are resampled in view 2 with probability α.

Content law. To test H3, we reweight how often each object-hue and scale value is sampled: uniform (the dataset's grid), or a discretised Gaussian or Laplace law with unit variance over a grid spanning ±2.5 sd. The images stay real; only their sampling frequencies change. Shape stays uniform.

Temporal tier. Clips of $T = 3$ frames from 3DShapes: content is fixed, and each nuisance factor takes a ±1 step per frame with probability α (hues wrap; orientation reflects). The predictor sees the concatenated embeddings of frames 1–2 and predicts the embedding of frame 3. We chose 3DShapes clips over Moving MNIST [34] so that the temporal tier shares the core tier's factors and probes.

Natural-image tier. CIFAR-10 [35]: two independent augmentations of one image. The eight augmentation parameters (crop centre, log scale, log aspect, flip, brightness, contrast, saturation) are the nuisance, independent across views; α scales augmentation strength. Content is probed through the class label on un-augmented test images.

3.2 Objective and bottlenecks

All arms share one encoder f_θ (three 4×4 stride-2 convolutions with 16, 32 and 32 channels, then an MLP $512 \to 256 \to d$, no normalisation layers so collapse stays visible), one MLP predictor g_ϕ ($d \to 256 \to d$), Adam with learning rate 10^{-3} (10^{-2} for the predictor), batches of 256 pairs and 2000 steps. With $z_v = f_\theta(x_v)$, the image-JEPA loss is

$$\mathcal{L} = \frac{1}{2}\left[\,\mathrm{MSE}(g_\phi(b_\sigma(z_1)), z_2) + \mathrm{MSE}(g_\phi(b_\sigma(z_2)), z_1)\right] + \lambda \cdot \frac{1}{2}\left[\,R(Z_1) + R(Z_2)\right],$$

where R is the arm's regulariser on the batch Z_v. The two bottlenecks are:

- **dimensional:** the embedding dimension $d \in \{3, 6, 12, 24\}$, below and above $k = 6$;
- **information:** $b_\sigma(z) = z + \sigma \cdot \mathrm{sd}_B(z) \odot \xi$, $\xi \sim \mathcal{N}(0, I)$, noise on the predictor input scaled by the detached batch standard deviation, so that the per-dimension signal-to-noise ratio is $1/\sigma^2$ whatever the scale of z. Probes use the clean z. Without the scaling, the EMA and unregularised arms could escape the noise by growing z.

3.3 Arms

1. **none**: $R = 0$ (collapse check).
2. **VICReg**: variance hinge plus 0.04 × covariance penalty, the 25:1 ratio of the original paper [4], with invariance played by the prediction loss.
3. **SIGReg**: the Epps–Pulley statistic of 256 random one-dimensional projections, computed by the official `lejepa` package [5].
4. **sliced KSD**: for random unit directions g, the one-dimensional IMQ Stein kernel on $t = g^T z$ with projected score $g^T \nabla \log p(z)$ (the $r = g$ variant of sliced KSD [17]), averaged over 32 directions resampled every step.

5. **KSD**: the U-statistic $\widehat{\mathrm{KSD}}^2 = \frac{1}{n(n-1)} \sum_{i \neq j} u_p(z_i, z_j)$ with Langevin–Stein kernel $u_p(x,y) = s(x)^T s(y)\, k(x,y) + s(x)^T \nabla_y k + s(y)^T \nabla_x k + \mathrm{tr}\ \nabla_x \nabla_y k$, score $s = \nabla \log p$ and IMQ base kernel $k(x,y) = (1 + \|x-y\|^2)^{-1/2}$ [14], [16].
6. **MMD**: unbiased MMD² against a fresh prior sample with Gaussian kernels $\sum_{h \in \{0.5,1,2\}} \exp(-\|x-y\|^2 / 2h^2 d)$.
7. **EMA**: no regulariser; the target is $\mathrm{LayerNorm}(f_\xi(x_2))$ under stop-gradient, with $\xi \leftarrow \tau\xi + (1-\tau)\theta$, as in V-JEPA. Without the target LayerNorm the BYOL-style variant collapsed in the pilot.
8. **oracle** (diagnostic only): KSD towards the rank-k target $\mathcal{N}(0, \mathrm{diag}(1_k, \varepsilon^2 1_{d-k}))$ with $\varepsilon = 0.3$, which uses the known k and is not a practical method.

The target law p is $\mathcal{N}(0, I_d)$ unless stated. For H3, the **matched priors** are product laws with unit-variance coordinates: Laplace, with score $-\tanh(x/0.1)/b$ (a smoothed sign, $b = 1/\sqrt{2}$), and uniform, whose score is undefined; we use the generalised-normal surrogate with shape 8 as a smooth stand-in, and draw exact uniform samples for MMD and NP-KSD.

The temporal tier runs EMA alone (the V-JEPA recipe), EMA + sliced KSD, EMA + SIGReg, sliced KSD alone and VICReg. The **autoencoder tier** uses the same encoder with a mirrored deconvolutional decoder and per-image squared reconstruction error, plus either a β-weighted KL to $\mathcal{N}(0, I)$ (β-VAE [29]), λ·KSD of the deterministic code batch towards the prior (KSD-AE), or λ·MMD (MMD-AE, as WAE-MMD [31]).

3.4 Metrics

Probes are fitted on frozen embeddings of 10,000 training-split images and evaluated on 5,000 held-out test-split images, both without view noise and drawn from the run's content law.

- **content R^2**: ridge-probe R^2 of the 6-dimensional content code, averaged over targets (CIFAR: logistic-regression class accuracy);
- **leakage R^2** (primary): ridge-probe R^2 of the 5-dimensional nuisance code (CIFAR: of the 8 augmentation parameters); lower is better. We report kNN ($k = 10$) R^2 beside it, because the pilot found leakage that only a nonlinear probe detects;
- **identifiability**: R^2 of the best scaled semi-orthogonal map from centred embeddings to the standardised content code. It is 1 for a rotated copy of the content and 0.74 for a random linear mix of it (validation);
- **effective rank**: exp of the entropy of the normalised singular values of the embedding matrix (the quantity RankMe uses [33]); meaningless under collapse, so read with the embedding standard deviation;
- **Gaussianity / fit to the prior**: the sliced Epps–Pulley statistic of held-out embeddings, and an NP-KSD statistic [21] of 500 embeddings against the run's target law;
- **loss stability**: coefficient of variation of the loss over the last 20% of steps.

3.5 Protocol

Matched tuning. Every tuned arm gets three levels, $\lambda \times \{1/5, 1, 5\}$ around a pilot centre (EMA: $\tau \in \{0.99, 0.996, 0.999\}$), on two validation seeds (100, 101). The level with the highest mean validation linear content R^2 (CIFAR: class accuracy) is selected; leakage and identifiability are never used, so the selection cannot favour any hypothesis. Core arms are tuned at the **base setting** (3DShapes, $d = 12 = 2k$, $\alpha = 1$, $\sigma = 0$, uniform law) and the d, σ and α slices reuse those levels; H3 arms are re-tuned at the Laplace law; each other tier is tuned at its own base. Table 7 lists the selections. Several land on the lowest level of their grid (Section 6).

Held-out runs and statistics. Each selected configuration is retrained on 5 held-out seeds (1–5), which were not used in the pilot or in tuning: 495 runs in 99 configurations, none with a non-finite loss. A λ-sensitivity slice reruns all three tuning levels of the six λ-tuned arms at the base on the same seeds (90 runs, 60 of them new). We report means with 95% percentile-bootstrap intervals over seeds (10,000 resamples). Contrasts between arms resample each arm's seeds independently, because different arms share no randomness; interactions are differences in differences. With five seeds the percentile bootstrap is somewhat narrow, so we read contrasts whose interval ends within about 0.01 of zero as weak.

Validation gates. Before any sweep, every regulariser was checked to be near zero on $\mathcal{N}(0, I)$ samples and clearly larger on collapsed or anisotropic ones (Table 8); SIGReg matched `lejepa` on the same batch and directions (relative

difference 0.0); all losses passed float64 gradient checks and unit tests; probes recovered ground-truth factors (content R^2 1.000 from rotated true factors; leakage R^2 −0.0005 from content alone); and the unregularised arm collapsed on real images. SIGReg's statistic is n times the Epps–Pulley statistic and has a positive mean under the null.

4 Results

Table 2 gives every arm at the base setting. The unregularised arm collapses (embedding standard deviation below 0.01 in every seed of every configuration), and all regularised arms and EMA learn content. Linear content R^2 stays modest because 3DShapes content is discrete (320 combinations) and no arm is asked to arrange it linearly; kNN content R^2 reaches 0.992 for KSD.

Table 2: Base setting (3DShapes, $d = 12$, $\alpha = 1$, $\sigma = 0$, uniform law): mean over 5 held-out seeds [95% bootstrap CI]. Leakage is the primary metric; "ident." is the orthogonal identifiability R^2.

arm	λ	content R²	kNN cont.	leakage R²	kNN leak.	ident.	eff. rank	emb. sd
none	—	0.043 [0.003, 0.086]	0.026	0.080 [0.005, 0.154]	−0.054	0.024 [0.002, 0.050]	6.8	0.000
VICReg	5	0.378 [0.329, 0.407]	0.570	0.673 [0.542, 0.772]	0.683	0.283 [0.219, 0.332]	8.7	1.020
SIGReg	0.05	0.419 [0.370, 0.467]	0.718	0.250 [0.218, 0.281]	0.505	0.268 [0.226, 0.310]	11.0	1.062
sliced KSD	1	0.558 [0.534, 0.578]	0.895	0.019 [0.008, 0.029]	0.130	0.336 [0.307, 0.363]	5.7	1.003
KSD	0.2	0.727 [0.695, 0.748]	0.992	0.002 [−0.001, 0.005]	0.191	0.576 [0.543, 0.595]	10.3	0.817
MMD	10	0.629 [0.602, 0.661]	0.887	0.012 [0.000, 0.026]	0.215	0.354 [0.301, 0.427]	9.9	1.110
EMA	τ = 0.999	0.432 [0.345, 0.526]	0.785	0.215 [0.137, 0.283]	0.347	0.259 [0.196, 0.328]	4.0	0.102
oracle	0.1	0.452 [0.399, 0.501]	0.680	0.451 [0.365, 0.560]	0.577	0.315 [0.228, 0.398]	7.8	0.575

4.1 H1: dimension and isotropic-Gaussian targets

H1: for $d > k$, isotropic-Gaussian regularisers increase nuisance leakage compared with VICReg and EMA. **Refuted under the protocol, and dependent on λ.**

At the tuned λ and $d = 12$, the Gaussian-target arms leak less than VICReg (0.673 [0.542, 0.772]): SIGReg 0.250 [0.218, 0.281], sliced KSD 0.019 [0.008, 0.029], KSD 0.002 [−0.001, 0.005], MMD 0.012 [0.000, 0.026]. All 8 Gaussian-arm – VICReg contrasts at $d \in \{12, 24\}$ are negative with intervals that exclude zero (8 of 8). Against EMA (0.215 [0.137, 0.283]), sliced KSD, KSD and MMD leak less at $d = 12$ and SIGReg is not different (+0.035 [−0.041, 0.113]); at $d = 24$, SIGReg (−0.057 [−0.169, 0.056]) and KSD (−0.005 [−0.146, 0.134]) are not different from EMA. Among these tuned-level contrasts, the one result in the direction H1 predicts is nonlinear: SIGReg's kNN leakage exceeds EMA's at every d (at $d = 12$: +0.158 [0.094, 0.217]).

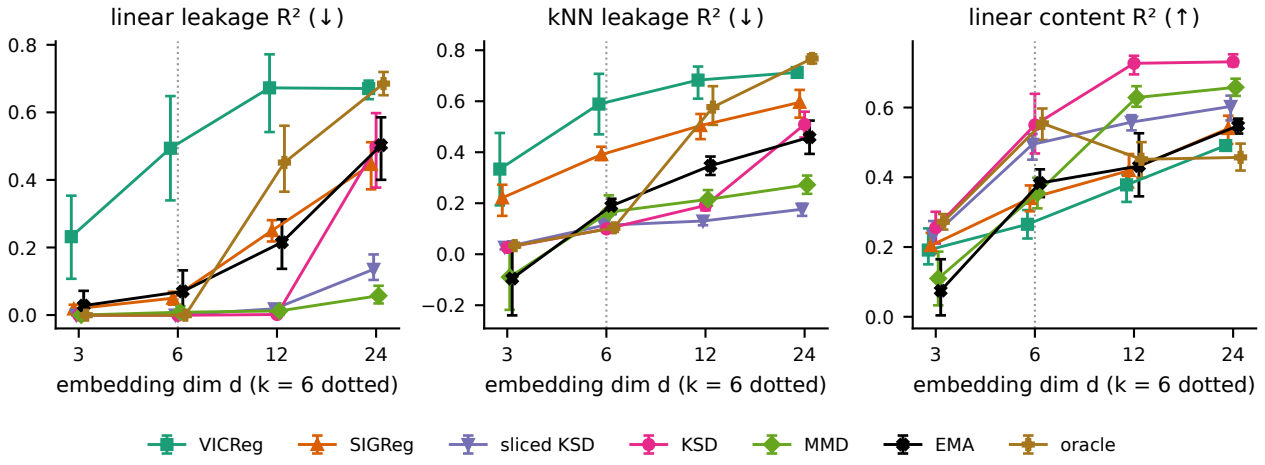


Figure 2: H1: nuisance leakage and content against embedding dimension d (base setting otherwise; tuned λ). Points are means over 5 held-out seeds with 95% bootstrap intervals; the dotted line marks $k = 6$. Every arm leaks more at $d = 24$ than at $d = 3$.

Figure 2 shows that every arm, not only the Gaussian ones, leaks more at $d = 24$ than at $d = 3$, though not monotonically: VICReg is flat from $d = 12$ to $d = 24$, and sliced KSD, KSD and MMD stay near zero up to $d = 12$ before rising.

From $d = 3$ to $d = 24$, VICReg goes from 0.232 to 0.671, SIGReg from 0.018 to 0.446, KSD from 0.000 to 0.498 and EMA from 0.028 to 0.503; sliced KSD (0.135) and MMD (0.057) stay lowest at $d = 24$. The oracle, whose target shrinks the $d - k$ extra directions to standard deviation 0.3, still leaks heavily for $d > k$ (0.451 at $d = 12$, 0.685 at $d = 24$): linear probes are scale-invariant, so shrinking a direction does not remove the nuisance it carries. At $d = 3$, EMA collapsed in 3/5 seeds and MMD in 2/5, so those two points mix collapsed and non-collapsed runs.

The weight λ decides the comparison. Tuning on validation seeds already showed leakage tracking λ, so we reran all three tuning levels on the held-out seeds (Figure 3). For every regulariser, VICReg included, the highest level leaks more than the lowest, with intervals excluding zero (6 of 6); for KSD the difference is +0.532 [0.458, 0.602] and for VICReg +0.499 [0.349, 0.596]. At five times its content-optimal level, each Gaussian-target arm leaks more than VICReg's least-leaky level (0.172 [0.142, 0.206]); for KSD at $\lambda = 1$ the gap is +0.210 [0.146, 0.266], and against EMA +0.166 [0.080, 0.254]. The selection rule (highest content R^2 on validation seeds) picked the lowest level for sliced KSD and KSD, whose held-out content falls with λ, and the middle level for the other four. Content is not monotone in λ for VICReg (0.265, 0.378, 0.315) or MMD (0.327, 0.629, 0.376), and for VICReg the rule picked a high-leakage level. Against VICReg's least-leaky level, the *tuned* SIGReg leaks more (+0.078 [0.032, 0.120]), a result in the hypothesised direction, while tuned sliced KSD, KSD and MMD still leak less (e.g. KSD −0.170 [−0.205, −0.140]). The "less than VICReg" part of the verdict therefore rests partly on VICReg's selected level. The verdict on H1 is a statement about content-tuned regularisers; under stronger distributional pressure the Gaussian arms do leak more than VICReg and EMA. More pressure buys leakage for every regulariser, so the effect is not specific to isotropic-Gaussian targets.

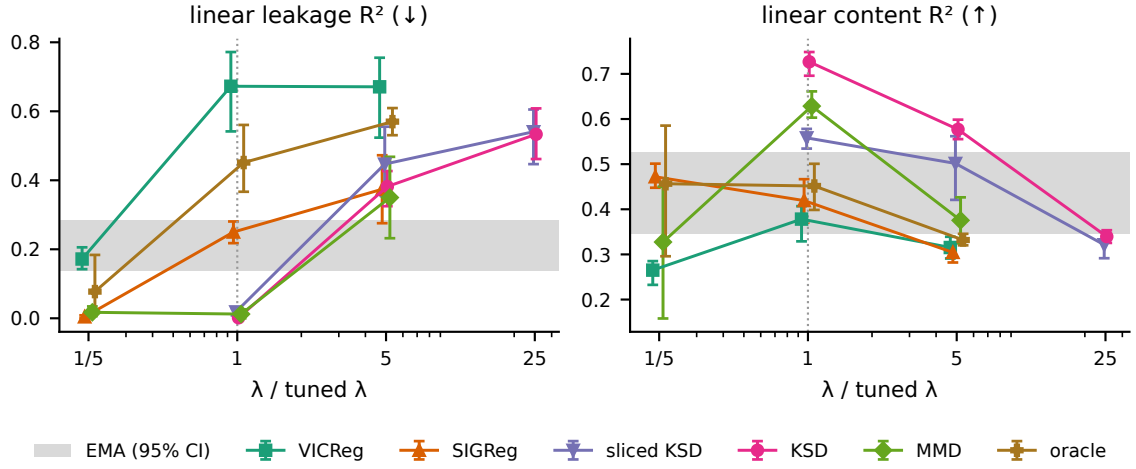


Figure 3: λ sensitivity at the base setting on held-out seeds: leakage and content at each arm's three tuning levels, plotted against λ divided by the tuned λ. The grey band is EMA's 95% interval.

4.2 H2: dimensional against information bottleneck

H2: the information bottleneck (σ) reduces leakage where the dimensional one (d) does not. **Refuted; the opposite holds.**

Narrowing d reduces linear leakage for all six arms ($\sigma = 0$, $d = 6$ against $d = 24$, intervals excluding zero): for example EMA −0.433 [−0.539, −0.322], KSD −0.499 [−0.599, −0.396] and MMD −0.049 [−0.081, −0.020]; for VICReg the evidence is weak (−0.177 [−0.338, −0.012]). Injected noise never reduces linear leakage (Figure 4). At $\sigma = 1$ against $\sigma = 0$, leakage rises for EMA, SIGReg, sliced KSD, KSD and MMD at both d, for example KSD at $d = 12$ +0.646 [0.596, 0.687] and MMD +0.405 [0.347, 0.469]; VICReg is unchanged (−0.103 [−0.243, 0.050]). Noise also lowers content R^2 for every arm. The only reduction is EMA's kNN leakage (−0.107 [−0.153, −0.060] at $d = 12$), and it comes with a larger loss of kNN content (0.785 ⟶ 0.244): the embedding degrades rather than filtering nuisance. EMA's loss is also unstable under noise (coefficient of variation of the loss 0.83 at $\sigma = 0.5$ and 0.58 at $\sigma = 1$ for $d = 12$, against 0.12 without noise).

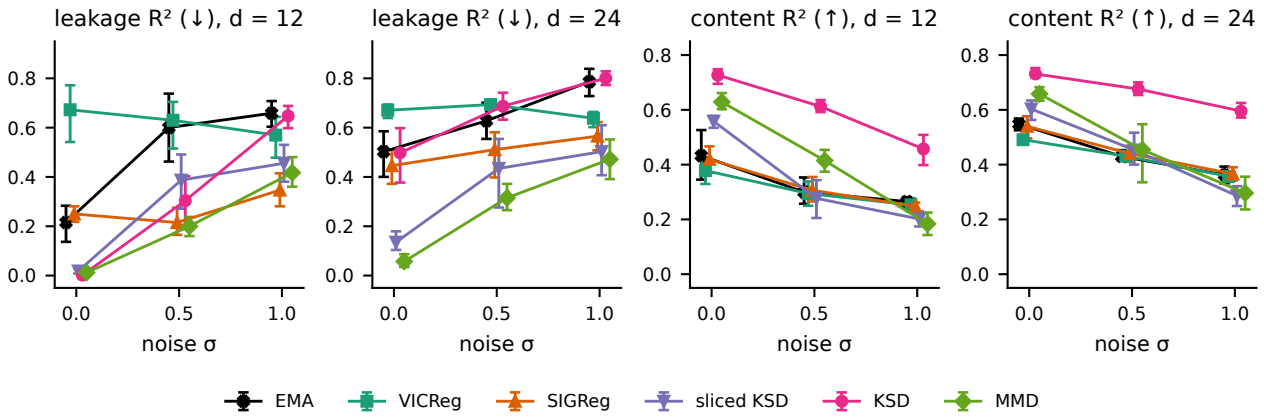


Figure 4: H2: leakage and content against injected noise σ at $d = 12$ and $d = 24$ (α = 1, uniform law, tuned λ). Noise never lowers linear leakage and lowers content for every arm.

4.3 H3: matched priors under non-Gaussian content laws

H3: with non-Gaussian content laws, KSD with a matched prior beats Gaussian-target regularisers on identifiability. **Refuted.**

Under the uniform law, matched-uniform KSD (0.554 [0.501, 0.598]) is not different from KSD towards $\mathcal{N}(0, I)$ (0.576 [0.543, 0.595]; difference −0.022 [−0.082, 0.036]). It beats SIGReg, sliced KSD, MMD and the oracle, but so does Gaussian-target KSD: the advantage belongs to full KSD, not to the matched prior. Under the Laplace law, matched-Laplace KSD (0.276 [0.235, 0.317]) is far below Gaussian-target KSD (0.772 [0.717, 0.813]; difference −0.496 [−0.558, −0.430]) and also below SIGReg (−0.082 [−0.157, −0.008], weak). Matched sliced KSD is below every Gaussian-target arm at both laws (Figure 5, Table 3). Full KSD towards $\mathcal{N}(0, I)$ has the best identifiability at every law, and its highest value is under the Laplace law; so do SIGReg and sliced KSD, while MMD and the oracle peak under the Gaussian law. This differs from Klindt et al., whose generalised-normal sweep of the latent law with LeJEPA has identifiability peaking at the Gaussian law [7]. The settings differ in ways we cannot separate: their latents are continuous with d equal to the latent dimension and views differ only by transition noise, whereas here the law reweights two factors of a discrete grid, $d = 12 > k$, nuisance is present, and the probe images are drawn from each law, so scores at different laws are measured on different populations.

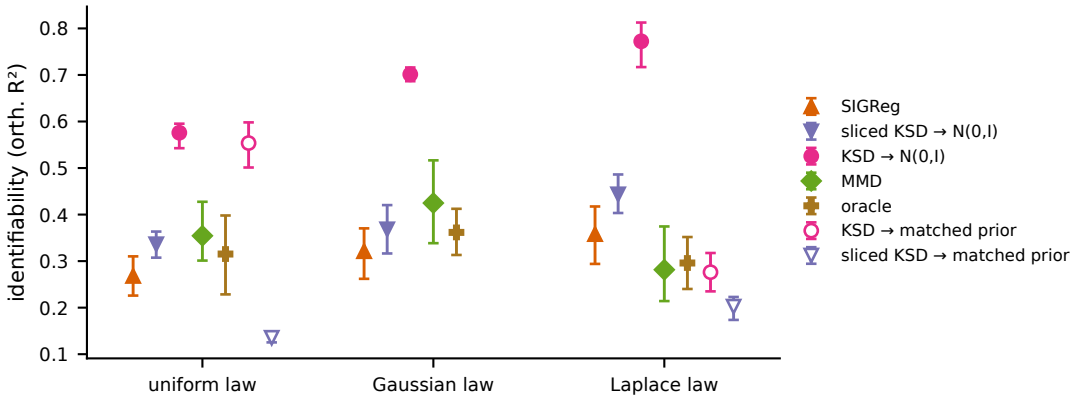


Figure 5: H3: orthogonal identifiability by content law (d = 12). Filled markers: Gaussian targets; open markers: KSD and sliced KSD with a prior matched to the content law (at the Gaussian law the matched prior is $\mathcal{N}(0, I)$).

Three caveats limit this verdict. Laplace-target KSD has little power at batch size 256: in validation it scored −0.0037 on an exact Laplace sample and 0.0040 on a Gaussian one (uniform target: 0.0075 against 3.40). The trained matched-Laplace embeddings do not reach their prior (NP-KSD 1.14 for KSD and 7.42 for sliced KSD, against 0.02 for Gaussian-target

KSD towards its own target). And the matched prior acts on all $d = 12$ coordinates while the law changes only two content factors.

Table 3: H3: identifiability (orthogonal R^2) by content law and target; mean [95% CI] over 5 held-out seeds.

arm (target)	uniform law	Gaussian law	Laplace law
SIGReg	0.268 [0.226, 0.310]	0.321 [0.262, 0.370]	0.358 [0.294, 0.417]
sliced KSD → N(0, I)	0.336 [0.307, 0.363]	0.368 [0.316, 0.420]	0.444 [0.403, 0.486]
KSD → N(0, I)	0.576 [0.543, 0.595]	0.701 [0.687, 0.716]	0.772 [0.717, 0.813]
MMD	0.354 [0.301, 0.427]	0.425 [0.339, 0.517]	0.281 [0.214, 0.374]
oracle	0.315 [0.228, 0.398]	0.362 [0.313, 0.412]	0.296 [0.240, 0.352]
KSD → matched prior	0.554 [0.501, 0.598]	(= N(0, I) row)	0.276 [0.235, 0.317]
sliced KSD → matched prior	0.135 [0.125, 0.145]	(= N(0, I) row)	0.202 [0.174, 0.223]

4.4 H4: EMA under nuisance

H4: EMA/stop-gradient behaves qualitatively differently from distributional regularisers under nuisance. **Refuted for the leakage response; its geometry does differ.**

Raising α from 0 to 1 lowers leakage or leaves it unchanged for every arm and raises content (Figure 6): EMA −0.458 [−0.537, −0.384], KSD −0.410 [−0.479, −0.332], sliced KSD −0.521 [−0.567, −0.455], MMD −0.336 [−0.376, −0.289], SIGReg −0.113 [−0.173, −0.042], VICReg −0.041 [−0.180, 0.085]. In differences of differences, EMA's response is not different from KSD's (−0.048 [−0.161, 0.055]) or sliced KSD's (+0.063 [−0.036, 0.155]), and EMA's drop is larger than VICReg's (−0.417 [−0.563, −0.260]), SIGReg's (−0.345 [−0.449, −0.246]) and MMD's (−0.122 [−0.214, −0.034]). The size of EMA's response thus differs from three of the five regularisers but lies inside their range, and its direction is shared by all; we read this as no *qualitative* difference. At α = 0.5 EMA leaks 0.548, close to VICReg (0.539), while SIGReg leaks 0.338 and sliced KSD, KSD and MMD leak 0.006, 0.001 and 0.006, so the middle level separates sliced KSD, KSD and MMD from EMA and VICReg, with SIGReg between, rather than EMA from all regularisers. EMA does differ in geometry: at α = 1 its embedding has standard deviation 0.102 and effective rank 4.0, against 5.7–11.0 for the regularisers.

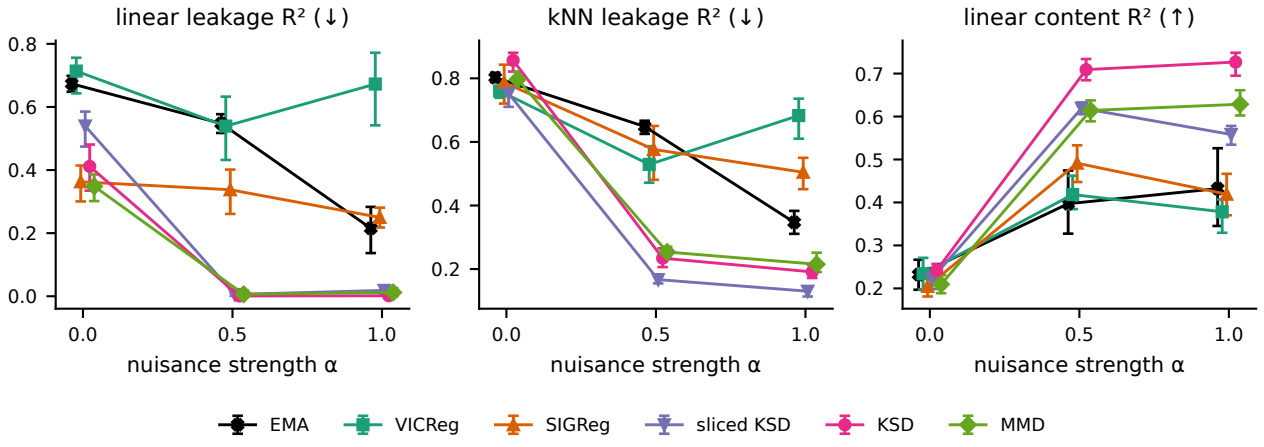


Figure 6: H4: leakage and content against nuisance strength α (d = 12, σ = 0). At α = 0 the two views are the same image, so nuisance is fully predictable from the other view.

4.5 H5: temporal tier

H5: with nuisance that moves over time, EMA + stop-gradient alone leaks less nuisance than any distributional regulariser alone, and adding sliced KSD to EMA leaves leakage unchanged within the CIs. **Refuted on both parts** (Table 4, Figure 7 left).

EMA alone leaks more than sliced KSD alone (+0.385 [0.262, 0.519]) and is not different from VICReg (+0.071 [−0.081, 0.230]; on kNN leakage EMA is higher, +0.212 [0.094, 0.333]). Adding sliced KSD to EMA raises leakage (+0.229 [0.110, 0.337]); adding SIGReg gives +0.121 [−0.005, 0.236]. Leakage is high for every temporal arm except sliced KSD alone (0.098 [0.033, 0.164]), plausibly because the target frame's nuisance is one ±1 step from the context, so it is predictable from the context and an encoder that keeps it lowers the prediction loss.

Table 4: Temporal tier (3DShapes clips, $T = 3$, $\alpha = 1$, $d = 12$). The EMA + X arms keep $\tau = 0.999$ and tune λ.

arm	λ	content R^2	leakage R^2	kNN leakage R^2	eff. rank
EMA	$\tau = 0.99$	0.338 [0.297, 0.384]	0.484 [0.381, 0.601]	0.553 [0.462, 0.644]	7.3
EMA + sliced KSD	5 ($\tau = 0.999$)	0.284 [0.272, 0.296]	0.712 [0.686, 0.741]	0.808 [0.776, 0.844]	12.0
EMA + SIGReg	0.05 ($\tau = 0.999$)	0.251 [0.229, 0.270]	0.605 [0.564, 0.655]	0.779 [0.774, 0.786]	12.0
sliced KSD	1	0.562 [0.510, 0.610]	0.098 [0.033, 0.164]	0.284 [0.232, 0.333]	7.0
VICReg	1	0.393 [0.354, 0.422]	0.413 [0.307, 0.518]	0.341 [0.274, 0.422]	4.3

4.6 H6: autoencoders against JEPA

H6: at matched d, reconstruction-based models leak more nuisance than every JEPA arm. **Confirmed.** All 21 contrasts (three autoencoders × seven JEPA arms, $d = 12$, linear leakage) are positive with intervals excluding zero (21 of 21), and so are 21 of 21 kNN contrasts (Table 5, Figure 7 right). The smallest gap is β-VAE against VICReg, +0.136 [0.028, 0.267]. The matched-prior question of H3 carries over as a null-to-negative result: under the Laplace law, the Laplace-prior KSD autoencoder has lower identifiability than the Gaussian-prior one (−0.121 [−0.204, −0.036]) and the same content R^2 (−0.011 [−0.081, 0.054]).

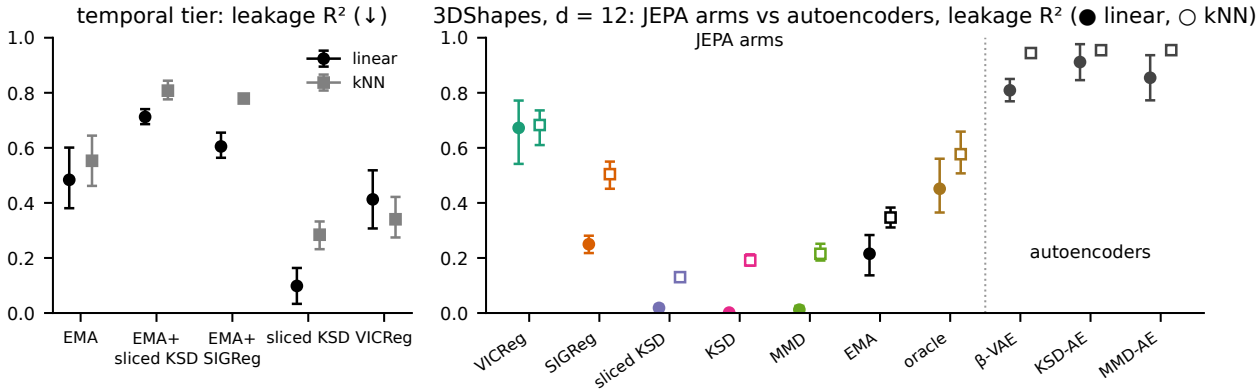


Figure 7: Left: temporal-tier leakage (linear and kNN). Right: leakage of the seven JEPA arms and three autoencoders at $d = 12$ on 3DShapes (filled: linear; open: kNN).

Table 5: Autoencoder tier (3DShapes, $d = 12$); λ / β tuned for content R^2.

model	λ / β	content R^2	leakage R^2	kNN leak.	ident.
β-VAE (KL to N(0, I))	0.2	0.324 [0.292, 0.350]	0.809 [0.769, 0.850]	0.944	0.140 [0.113, 0.168]
KSD-AE ⟶ N(0, I)	10	0.366 [0.313, 0.420]	0.912 [0.846, 0.977]	0.955	0.203 [0.162, 0.243]
MMD-AE ⟶ N(0, I)	20	0.337 [0.270, 0.393]	0.855 [0.773, 0.937]	0.955	0.167 [0.121, 0.208]
KSD-AE ⟶ N(0, I), Laplace law	2	0.485 [0.431, 0.531]	0.957 [0.908, 0.982]	0.966	0.309 [0.263, 0.364]
KSD-AE ⟶ Laplace prior, Laplace law	2	0.474 [0.422, 0.505]	0.935 [0.872, 0.979]	0.963	0.188 [0.123, 0.256]

4.7 Natural images (CIFAR-10)

The CIFAR-10 tier carries no hypothesis (Table 6). Every regularised arm reaches similar class accuracy, KSD highest (0.357 [0.350, 0.365]) and EMA lowest of the trained arms (0.299 [0.281, 0.316]); the unregularised arm collapses. Linear leakage of the augmentation parameters is small for every arm (0.018 to 0.049, EMA highest); Gaussian-target arms differ from VICReg by at most a few hundredths. With augmentation-defined nuisance and the class as content, the isotropic regularisers do not leak nuisance linearly to any notable degree. The oracle is omitted here because CIFAR has no known k.

Table 6: CIFAR-10 tier (α = 1, $d = 12$): class accuracy and linear leakage of the eight augmentation parameters.

arm	λ	class accuracy	augmentation leakage R^2	eff. rank	emb. sd
none	—	0.115 [0.104, 0.128]	−0.021 [−0.061, −0.001]	6.8	0.000
VICReg	5	0.333 [0.324, 0.343]	0.022 [0.014, 0.028]	9.7	1.116
SIGReg	0.05	0.332 [0.324, 0.340]	0.039 [0.031, 0.045]	11.7	0.923
sliced KSD	5	0.348 [0.341, 0.356]	0.036 [0.027, 0.044]	10.6	0.899
KSD	0.2	0.357 [0.350, 0.365]	0.018 [0.012, 0.023]	10.7	0.616
MMD	10	0.341 [0.328, 0.354]	0.020 [0.013, 0.027]	11.2	0.913
EMA	τ = 0.999	0.299 [0.281, 0.316]	0.049 [0.042, 0.057]	4.7	0.073

5 Discussion: dimensional against information bottleneck

The results fit one reading, which we offer as an interpretation rather than a tested claim. A distributional regulariser asks for d dimensions of spread. When the shared content provides fewer ($d > k$), the cheapest way to meet the demand is to spend the spare directions on whatever else varies in the image, and on 3DShapes that is the nuisance. Three manipulations change how hard the demand presses, and for the distributional regularisers all three move leakage the way this reading predicts: larger d leaves more spare directions (Figure 2); larger λ weights the demand more heavily (Figure 3); and the scaled noise of H2 makes every direction less useful for prediction, which weakens the pull of the prediction loss towards content relative to the regulariser (Figure 4). Only shrinking d removes room for nuisance. EMA has no explicit demand, and its embedding stays low-rank, but it still leaks once d exceeds k: in a predictive objective without a spread constraint nothing pushes nuisance out of the spare directions either.

On this reading the dimensional bottleneck is a filter and the information bottleneck, as we implemented it, is not. The noise here is scaled to the batch standard deviation of each coordinate, so the encoder cannot raise the signal-to-noise ratio of content relative to nuisance by rescaling; an information bottleneck that charges per bit of $I(z; x)$, such as a learned-variance KL term, or noise fixed in absolute scale as in the analysis of Arnez et al. [27], might behave differently. We did not test those. A second qualification: the predictor reads the noisy z while the probes read the clean z, so the noise can also change what the encoder writes into dimensions the predictor learns to ignore.

The H3 result runs against the motivation for matched targets. Two explanations are compatible with our data and we cannot separate them: KSD towards a Laplace target has little power at our batch size, so the matched arm receives a weak and noisy training signal; and the matched prior constrains all twelve coordinates while only two content factors follow the law. That full KSD towards $\mathcal{N}(0, I)$ is best at every law is consistent with KerJEPA including unsliced KSD among its variants [6] and suggests that, at this scale, the quality of the training signal matters more than the shape of the target.

Finally, the λ-sensitivity slice is the result we would most want a practitioner to see. A study that fixed λ for each regulariser at a conventional value could reach the opposite verdict on H1 from ours, and both would be correct for their λ. Leakage comparisons between regularisers are only meaningful together with the rule that set their strengths.

6 Limitations

- **Rendered core data.** 3DShapes is rendered, with a small discrete factor grid; content R^2 is bounded by its discreteness, and its nuisance (colours, orientation) is simpler than real nuisance.
- **Augmentation-defined nuisance.** On CIFAR-10 the nuisance is our augmentation, not a property of the scene, so its leakage measures invariance to our own transformations.
- **Small scale.** A three-layer encoder at 32×32, 2000 steps and $d \leq 24$. All runs trained on one laptop GPU, six at a time (28.4 hours of summed per-run training and evaluation time). Larger encoders and longer training may change every effect.
- **Tuning.** λ was tuned only at the base setting and reused across d, σ and α, so part of the d and σ effects may come from a λ that is no longer optimal there. Several selected levels sit at the lowest end of their grid (sliced KSD, KSD, matched-uniform KSD; SIGReg and oracle at the Laplace law; several autoencoders), so lower λ might score higher still. The validation selection is noisy: SIGReg's held-out content R^2 is higher at λ = 0.01 (0.472) than at the selected 0.05 (0.419).

- **Statistics.** Five seeds per configuration, unpaired across arms, with percentile-bootstrap intervals that are somewhat narrow at $n = 5$. The EMA arm is the least stable across seeds.
- **The oracle** uses the true content dimension and is a diagnostic only.
- **One information bottleneck** (batch-scaled noise on the predictor input) among several possible.
- **Literature coverage.** Novelty claims are relative to 35 reviewed works found by a targeted search; 2026 preprints are not exhaustively covered.

7 Future work

The natural next steps are data with richer nuisance and known factors — Causal3DIdent [25] and MPI3D-real — and natural images with a held-out nuisance label, such as ImageNette with recorded augmentation; real video for the temporal tier; an information bottleneck with a learned rate; and λ tuned at every point of the d and σ slices.

8 Reproducibility statement

All code, configurations and per-run results are in the study repository. `REPRODUCE.md` gives the exact commands, seeds, environment and hardware. Every run is keyed by a hash of its full configuration; the tables in `results/` are generated by `stein.analyze`, and every number, table and figure in this paper is generated from them by `stein.report` and inserted at render time — no number in the text is typed by hand. Seeds are fixed. As a check, we retrained the 25 temporal-tier runs from scratch in an empty run cache on the same machine: 25 of 25 reproduced every metric bitwise. MPS kernels are not guaranteed to be deterministic across hardware or software versions, so reruns elsewhere may differ slightly. The study (experiments, analysis and this draft) was carried out by an AI agent (Claude) under a protocol and hypotheses fixed before the sweep, with its own validation gates.

9 Appendix

Table 7: Selected levels at the 3DShapes base setting (validation seeds 100, 101; criterion: mean linear content R^2). Matched-prior, Laplace-law, temporal, CIFAR and autoencoder selections are in `results/tuning_table.md`.

arm (3DShapes base)	selected level	validation content R^2
VICReg	5	0.391
SIGReg	0.05	0.537
sliced KSD	1	0.538
KSD	0.2	0.705
MMD	10	0.610
oracle	0.1	0.438
EMA	τ = 0.999	0.505

Table 8: Validation gate: regulariser value on 256 samples in $d = 12$, mean of 5 draws, for $\mathcal{N}(0, I)$ and four degenerate distributions ("collapsed" = $1 + 0.01\,\mathcal{N}$; anisotropic sd 3/1/0.1; mean shifted by 0.5). VICReg is invariant to a mean shift by design.

regulariser	N(0, I)	collapsed	rank 1	anisotropic	shifted mean
VICReg	0.018	0.986	0.716	0.248	0.020
SIGReg	0.886	278.1	80.8	33.1	31.2
sliced KSD	−0.001	2.096	0.663	0.167	0.159
KSD	−0.001	23.9	4.725	0.365	0.629
MMD	0.000	0.597	0.213	0.060	0.050

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